

KodetoCareer — PRD (Updated Sections Only)

Version 2.0 | Free Stack Corrections

Only the sections with paid service references are updated below. All other PRD content (modules 1–9 functional requirements, non-functional requirements, MVP scope, success metrics) remains identical to PRD v1.

CHANGED SECTION: Module 2.2 — Video Management

FR-2.2.1 (Updated)

Uploaded videos shall be stored on **Cloudflare R2** and served via **Cloudflare CDN** (free, zero egress cost). ~~AWS S3 and CloudFront~~ are not used.

FR-2.2.2 (Unchanged)

The system shall support two video source types: direct upload (MP4 up to 2GB) and embedded URL (YouTube, Vimeo, Google Drive).

FR-2.2.3 (Updated)

~~Direct upload videos shall be transcoded to adaptive bitrate (480p, 720p, 1080p) using AWS MediaConvert.~~

Replaced with: Direct-upload videos are stored as original MP4 on Cloudflare R2 and served directly from the Cloudflare CDN — no server-side transcoding for MVP. For large course videos, trainers are encouraged to upload to YouTube (unlisted) and paste the embed URL, which provides adaptive quality delivery for free.

FR-2.2.4 (Updated)

Video player must support: playback speed control (0.5x to 2x), fullscreen, and picture-in-picture. ~~Quality selection~~ is not available for direct-upload MP4s in the MVP (no transcoding). YouTube/Vimeo embeds provide their own quality selector.

FR-2.2.5 (Unchanged) — Video watch progress persisted, resume from last position.

FR-2.2.6 (Unchanged) — Videos not downloadable by students (stream-only via CDN).

FR-2.2.7 (Unchanged) — Admin panel shows per-video analytics.

CHANGED SECTION: Module 8 — Communication System

FR-8.1 (Unchanged) — Announcements can be sent to any scope.

FR-8.2 (Unchanged) — Notifications appear in-app and trigger push notifications.

FR-8.3 (Updated)

The system shall send automated **email** notifications via **Resend** (free, 3,000/month) for: new lesson available, quiz deadline approaching, assignment graded, certificate ready, attendance below threshold. ~~AWS SES~~ is not used.

FR-8.4 (Updated)

~~A bulk SMS option (via Twilio or MSG91) shall be available for critical communications.~~

Replaced with: SMS is not included in the MVP. All OTPs and critical alerts are delivered via email. SMS (Twilio or MSG91) shall be added as an optional feature in Version 2 once the platform generates revenue. This eliminates ~\$1,500/month in recurring SMS cost during the startup phase.

FR-8.5 (Unchanged) — WhatsApp notification integration remains a Version 2 feature.

CHANGED SECTION: Non-Functional Requirements — Data Backup

Original:

~~Data backup frequency: Daily automated backup to separate AWS region~~

Updated:

Data backup is handled automatically by Neon.tech's built-in point-in-time recovery (free tier: 7 days). Cloudflare R2 provides 99.999999999% durability. No manual backup configuration required. No additional cost.

SUMMARY OF ALL SERVICE CHANGES IN PRD

Original | Updated | Cost Impact

AWS S3 + CloudFront | Cloudflare R2 + CDN | \$4,000/month → \$0

AWS MediaConvert (transcoding) | Removed from MVP; direct MP4 + YouTube embed | \$3,000/month → \$0

AWS SES (email) | Resend (free 3,000/month) | \$500/month → \$0

MSG91 / Twilio (SMS) | Email OTP only; SMS in v2 | \$800/month → \$0

AWS RDS backup | Neon.tech built-in | Included in \$0

End of PRD Updated Sections

All other PRD content is unchanged from v1.